2026-10-14 | VS&A Daily |

When a stiffer panel lets more sound through



NASA Langley’s SALT anechoic receiving room, where sound transmitted through test panels was measured. Photo: NASA, Figure 10, Development of Quiet Honeycomb Panels (December 2009) [1].

A good structural decision can change the acoustic problem.

A lightweight sandwich panel is attractive for a familiar reason: separating its face sheets buys bending stiffness without filling the entire thickness with metal. But a panel that resists a static push efficiently can also become an efficient path for sound. Structural stiffness and acoustic insulation are not interchangeable measures of quality.

NASA’s honeycomb-panel work gives that warning a concrete setting. A 2006 study reported that, above 500 Hz in its test range, a single aluminum face sheet had greater measured transmission loss than a much heavier honeycomb sandwich. The research traced the behavior through the panel’s propagating waves, not mass alone [2].

Today’s objective is to recognize acoustic coincidence, estimate where it begins, and explain why changing stiffness can move the vulnerable band in the opposite direction from a familiar structural resonance. We will use an ideal plate calculation to expose the mechanism, then return to the limits that matter in a design review.

The distinction is worth keeping close: a natural frequency describes a standing vibration pattern. Coincidence describes a wavelength match between a traveling structural wave and the spatial pattern of an acoustic load.

Sound has a footprint along the panel

Consider an oblique plane sound wave approaching a flat panel. Its pressure does not rise and fall everywhere on the panel at once. The phase sweeps along the surface. The question is whether that moving pressure pattern can keep pace with a free bending wave.

Let θ be the incidence angle from the panel normal, c the sound speed, and ω = 2πf. The acoustic wavenumber is $k_0 = \omega/c$; the part parallel to the panel is:

$$k_{\parallel} = k_0 \sin\theta$$

The corresponding trace wavelength is longer than the wavelength in air. At normal incidence, the ideal pressure field is spatially uniform along an infinite panel. As incidence approaches grazing, its tangential wavenumber approaches k_0. Confusing the normal-based angle with an angle measured from the surface reverses this geometry.

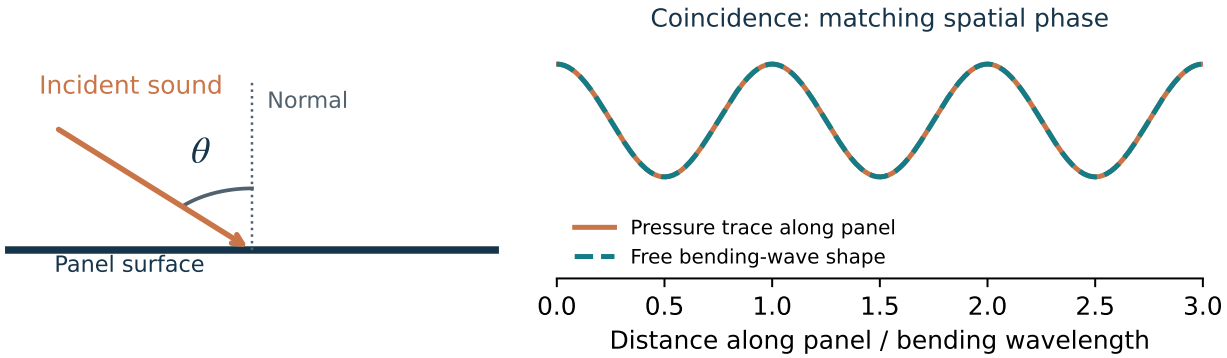


Figure 1. Original conceptual diagram. Coincidence matches wavelengths and spatial phase; the coincident traces are normalized, not a claim about equal pressure and displacement amplitudes or their temporal phase.

For a homogeneous, isotropic thin plate, flexural rigidity D has units N·m and areal mass μ has units kg/m². Neglecting fluid loading, shear deformation and rotary inertia, its free bending-wave dispersion relation gives:

$$k_b = \left(\frac{\mu\omega^2}{D}\right)^{1/4}, \qquad c_b = \frac{\omega}{k_b}$$

Unlike sound in the assumed nondispersive air, the bending phase speed increases with the square root of frequency. Below the critical frequency, it is too slow to match any propagating plane wave in the air. At the threshold it matches grazing incidence; above it, matching becomes possible at less oblique angles [3].

“Supersonic” here describes bending-wave phase speed relative to the surrounding fluid. It does not mean the panel material particles travel at hundreds of metres per second.

A stiffness increase moves the crossing downward

Equating k_b to the grazing acoustic wavenumber gives the critical frequency f_c. Matching an arbitrary oblique angle then gives f_{coin}. The critical frequency is the lowest member of that family, not a universal single resonance peak [3].

$$f_c = \frac{c^2}{2\pi}\sqrt{\frac{\mu}{D}}, \qquad f_{\mathrm{coin}}(\theta) = \frac{f_c}{\sin^2\theta}$$

Use an illustrative 2 mm aluminum-like plate: E = 70 GPa, density 2700 kg/m³, Poisson ratio ν = 0.33, and c = 343 m/s. For this homogeneous plate, $D = Eh^3/[12(1-\nu^2)]$ and $\mu = \rho h$. Thus D = 52.37 N·m, μ = 5.40 kg/m², and f_c = 6.01 kHz. These are selected inputs, not properties fitted to the NASA panels.

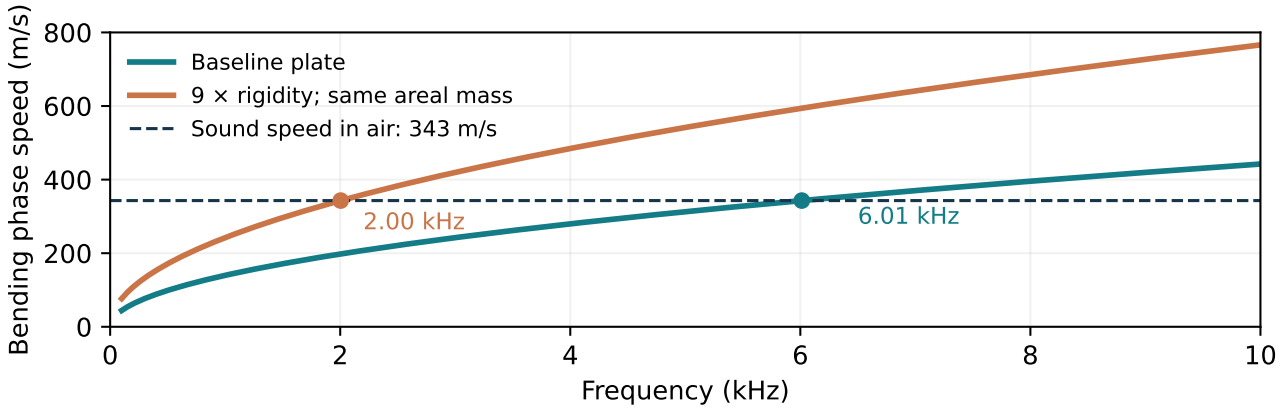


Figure 2. Original ideal-plate calculations. Multiplying D by nine at fixed μ raises bending-wave speed and lowers the crossing frequency by three. No transmission-loss amplitudes are predicted by this plot.

Illustrative change	μ (kg/m²)	D (N·m)	fc (kHz)	fcoin at 45°
2 mm baseline	5.40	52.37	6.01	12.03 kHz
9 × D; unchanged μ	5.40	471.33	2.00	4.01 kHz
4 mm homogeneous plate	10.80	418.96	3.01	6.01 kHz

Table 1. Original calculations. The fixed-mass stiffness change is a hypothetical effective plate, not a design prescription for a real sandwich.

For the stiffer fixed-mass case, a 4.01 kHz tone can match at 45°. For the baseline it is below critical, so no real incidence angle satisfies the infinite-plate matching condition. Doubling homogeneous thickness also lowers critical frequency: rigidity grows as h^3 while areal mass grows only as h.

Why the mass-law answer is not the whole answer

Transmission loss compares incident and transmitted acoustic powers: more loss means better isolation. Away from elastic effects, a limp mass is a useful baseline. For normal incidence, equal air on both sides and no leakage, the ideal expression is [3]:

$$\mathrm{TL}_{\mathrm{mass}} = 10\log_{10}\left[1+\left(\frac{\pi f\mu}{\rho_0 c}\right)^2\right]$$

Here ρ_0 is air density, distinct from the material density. At 1 kHz with $\rho_0 = 1.20$ kg/m³, our 2 mm plate's limp-mass estimate is 32.3 dB. Doubling thickness raises that ideal estimate to 38.3 dB. The familiar gain of roughly 6 dB follows from doubling areal mass when the bracketed mass term dominates.

That is not a prediction of a finite panel's measured isolation at 1 kHz. It is a reference calculation with no bending stiffness, mounting modes, leaks or flanking paths. Nor is a normal-incidence value interchangeable with a diffuse-field result, which includes oblique waves.

Near coincidence, the elastic panel can follow a matching pressure pattern efficiently. Bending stiffness and inertia partly cancel in the forced-wave dynamic stiffness; damping and radiation loading then matter greatly. Knowing the critical frequency locates a mechanism. It does not determine the depth or width of the transmission-loss dip. Even two panels with the same crossing frequency can have different losses, edge constraints and radiation impedances, so the crossing alone cannot rank their acoustic performance.

NASA's researchers changed the wave behavior by selectively recessing honeycomb core and reinforcing the recesses. For their Recessed 25-152 configuration, measurements showed a 3-11 dB transmission-loss improvement over 170-1000 Hz. The panel was 5.1% heavier, while static deflections at the evaluated recessed locations were roughly twice the baseline values [2]. This was an acoustic-structural trade, not a free benefit.

Do not transfer those gains to an arbitrary flight panel. Real sandwiches have core shear, face-sheet bending and direction-dependent properties; the 2006 study explicitly analyzed those wave families. Its model-to-test comparisons also acknowledged uncertainty in properties, boundaries and procedures [2]. Our single-rigidity formula deliberately omits that complexity.

Finite edges relax the ideal traveling-wave picture, so subcritical panels still transmit and radiate sound. Curvature, membrane stress and attachments introduce other mechanisms. Fluid loading may alter structural dispersion; a vacuum-facing panel does not have the same transmission problem as an air-to-air partition.

Design around the band that matters

Start a review with the source spectrum and the receiving-side requirement. A broadband cabin-noise objective and a narrow equipment tone may reward different changes. Place the estimated critical region on that spectrum before assuming that a thicker skin or a stiffer core is an acoustic improvement.

Then ask which waves the proposed model can represent. A thin isotropic plate may be enough for screening; an orthotropic or shear-flexible sandwich needs appropriate dispersion relations. Carry credible stiffness, mass and fluid-property ranges through the crossing calculation. A precise critical frequency from uncertain effective properties is false precision.

Validate the transmission path, not just a local vibration level. The SALT research used a reverberant source room and an anechoic receiving room separated by the test panel, measuring incident and transmitted acoustic quantities [1]. In a vehicle, seals, joints and flanking structure can bypass a good panel. A quieter local accelerometer channel alone cannot demonstrate improved cabin isolation.

For design changes, compare damping, local compliance and layered constructions against strength, stability, durability and mass. Added layers can introduce their own coupled resonances. This lesson offers a screening argument, not a qualification criterion: the cited NASA publications are research and technical guidance, not universal acceptance requirements.

Three takeaways

1. Coincidence is a spatial wavelength match, distinct from an individual standing-wave mode.
2. Higher bending rigidity per unit areal mass lowers the ideal critical frequency.
3. Crossing frequency identifies risk; transmission-loss magnitude still needs the actual panel, damping, loading and paths.

Reflection. At fixed areal mass, rigidity increases fourfold. What happens to critical frequency, and can you promise better sound isolation?
Answer. Critical frequency halves. No isolation promise follows: the coincidence region may move into a more important source band.

Sources, scope and visual credits

[1] D. L. Palumbo and J. Klos, Development of Quiet Honeycomb Panels, NASA/TM-2009-215954, December 2009. Section II and Figure 10: test method and opening photograph. NASA report.

[2] F. W. Grosveld, D. L. Palumbo, J. Klos and W. D. Castle, Finite element development of honeycomb panel configurations with improved transmission loss, INTER-NOISE, December 3-6, 2006. Figures 2 and 15; Table 3. Original study.

[3] J. R. Goodman and F. W. Grosveld (eds.), Acoustics and Noise Control in Space Crew Compartments, NASA/SP-2015-624, October 2015. Chapter VIII, pp. 26-28: critical frequency and mass law. NASA technical reference.

Figures 1-2 and Table 1 are original analytical illustrations. The NASA photograph is cropped from the report page; no experimental curves or flight telemetry have been reconstructed.